AI-Powered Outbound Calls (Arabic)

1. Use Case 1: AI-Powered Outbound Calls (Arabic)

1. Objective

- **Ensure** customers have correctly followed procedures reviewed on their prior inbound call.
- **Boost** First Call Resolution (FCR) KPI by proactively verifying compliance.
- **Automate** routine follow-up calls, freeing live agents for complex issues.
- **Generate** a documented “First Call Resolutions” report for quality assurance.

2. Functional & Technical Requirements

1. **Arabic Speech Recognition & Synthesis**
   - High-quality STT in Egyptian (and Modern Standard) Arabic.
   - Natural-sounding TTS in Arabic dialect(s).
2. **Call Orchestration**
   - Integration with your telephony provider (e.g., Twilio, Vonage, or on-prem PBX).
   - Retry logic, fall-back to human agent if AI fails.
3. **Conversation Logic**
   - Dynamic script that pulls in details from the previous inbound call (e.g., ticket ID, procedure steps).
   - Branching dialogs based on customer responses (yes/no, uncertain, request live agent).
4. **Data & Reporting**
   - Real-time logging of call results, durations, and resolutions.
   - Automatic generation of the “First Call Resolutions” article/report.

3. Proposed Architecture

```
scss
CopyEdit
[CRM / Inbound Call Records]
            │
            ▼
  [Call Scheduler & API] ──▶ [AI Call Engine]
           │                    • STT (Arabic Whisper)
           │                    • NLU & Dialogue (GPT-4)
           │                    • TTS (Arabic Neural Voices)
           ▼
 [Telephony Provider API] ──▶ [Customer Phone]
           │
           ▼
[Logging & Reporting DB] ──▶ [FCR Dashboard / “First Call Resolutions”
Report]
```

4. Workflow Steps

1. **Fetch** list of customers flagged for follow-up (e.g., incomplete steps).
2. **Schedule** outbound call slot via telephony API.
3. **Dial** customer, play dynamic TTS greeting in Arabic:

   > “صباح الخير، معك نظام المتابعة من شركة XYZ. نتأكد الآن من استكمال الإجراءات التي ناقشناها في مكالمتك الأخيرة…”

4. **Listen** to customer’s responses, transcribe via STT.
5. **Validate** step completion; if “نعم”، mark as resolved; if “لا” or “غير متأكد”، prompt further assistance or transfer to live agent.
6. **Log** call outcome and duration.
7. **Compile** resolved calls into the “First Call Resolutions” article—auto-formatted for your quality team.

5. Key Success Metrics

- **FCR Rate**: Target ≥ 90% on first outbound follow-up.
- **Call Completion**: % of calls fully handled by AI without handoff.
- **Average Handle Time**: Time per follow-up call vs. live-agent baseline.
- **Report Accuracy**: % of AI-generated resolutions matching manual audit.

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2. Use Case 2: Arabic Chatbot Integrated with Internal Knowledge Base

1. Objective

- **Empower** agents with instant, accurate answers drawn from your proprietary documentation.
- **Reduce** average handle time (AHT) and training ramp-up by having an “always-on” Arabic assistant.

2. Requirements

1. **RAG (Retrieval-Augmented Generation)** setup to query your internal KB (PDFs, Wiki, SOPs).
2. **Arabic Language Understanding** for dialect nuance and formal registers.
3. **Agent UI Integration** (e.g., widget in contact-center desktop).
4. **Access Controls & Audit Trail** to secure proprietary data.

3. Proposed Architecture

```
scss
CopyEdit
[Internal KB Storage]
    ├─ Documents (PDF / Word / Wiki)
    └─ Structured Data (DB)
             |
   (1) Preprocessing & Embeddings ──▶ [Vector DB (e.g., Pinecone)]
             |
   (2) Agent Query ──▶ [Chat Interface]
             |           • LLM (GPT-4 Turbo)
             |           • RAG: retrieves top K docs
             ▼
      [Answer + Source Snippets] ──▶ [Agent Desktop / Chat UI]
             |
      [Logging / Feedback Loop]
```

4. Workflow Steps

1. **Ingest & Embed** internal docs nightly (convert to text, chunk, embed in vector DB).
2. **Agent Types** a query in Arabic:

   “ما هي خطوات تحديث بيانات العميل في النظام الداخلي؟”

3. **RAG Pipeline** retrieves relevant passages.
4. **LLM** crafts a concise, cited answer in Arabic, e.g.:

   “يمكنك تحديث بيانات العميل عبر الدخول إلى… (المصدر: قسم العمليات، صفحة 12)”.

5. **Agent** reviews, shares with customer, or asks follow-up.
6. **Feedback** (thumbs-up/down) feeds back into retraining / prompt-tuning.

5. Success Metrics

- **Time-to-Answer**: Average latency from query to answer (< 2 seconds).
- **Agent Satisfaction**: Surveys on answer accuracy (≥ 4.5/5).
- **Usage Rate**: % of inquiries handled by bot vs. manual search.
- **Knowledge Coverage**: % of KB with up-to-date embeddings.